Zehao Fan

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Education

Rensselaer Polytechnic Institute, Troy, NY, USA *Aug. 2024 – May 2029*
Ph.D. Student in Electrical, Computer, and Systems Engineering (Advisor: Prof. Liu Liu)

University of Science and Technology of China (USTC), Hefei, China *Sept. 2020 – Jun. 2024*
Bachelor of Engineering in Computer Science

Skills

Programming Languages:	Python, C/C++, Java, Verilog/SystemVerilog, Assembly Language, Shell, SQL
Frameworks & Tools:	PyTorch, Anaconda, Hugging Face, Docker, Git, MATLAB, Xilinx Vivado
Areas of Expertise:	Large Language Models (LLMs), Computer Architecture, Processing-in-Memory (PIM), Near Data Processing (NDP), Mixture-of-Experts (MoE), Algorithm Design and Memory Systems

Research Experiences

Project 1: STARC: Selective Token Access with Remapping and Clustering for Efficient LLM Decoding on PIM Systems *Oct. 2024 – Aug. 2025*
Advisor: Prof. Liu Liu
Keywords: PIM, LLMs, Sparse Attention, Hardware–Algorithm Co-design **[Code: EPIC-RPI/STARC]**
Summary: Proposed **STARC**, a key-value (KV) clustering-based data mapping scheme that bridges dynamic attention sparsity and the row-level access granularity of PIM. Developed an in-situ cosine K-means KV clustering method using existing HBM-PIM primitives to avoid GPU offloading of KV vectors for clustering and achieve hardware-aware token organization. In HBM-PIM simulation, STARC achieves up to **93% / 92% lower attention-layer latency and energy** than full attention and **78% / 65%** reductions over token-wise sparsity, with comparable model accuracy. For end-to-end decoding, it delivers **25–48% / 34–56% lower latency and energy** than full attention, and **13–21% / 11–18%** reductions over token-wise sparsity across LLaMA-7B, Mistral-7B, and GPT-13B.

Project 2: CAMEO: Context-Aware Mixture-of-Experts Orchestration for Inference *Aug. 2025 – Aug. 2026*
Advisor: Prof. Liu Liu
Keywords: MoE, GPU Offloading, NDP, Mixed-Precision Quantization, System Design
Summary: Designed **CAMEO**, a context-aware MoE orchestration framework for both GPU-offloading and GPU–NDP systems. Leveraged prefill-stage expert activation counts and routing-score sums to construct a **once-per-sequence** decoding plan that pins important experts in GPU HBM in FP16. Developed an expert placement module and a per-expert bitwidth selector that combines context-specific importance with an offline quantization-loss table to assign **1–4-bit replicas** to the remaining experts, reducing host–GPU transfers and NDP-local weight access and computation. Across Mixtral-8×7B, Phi-3.5-MoE, DeepSeek-MoE-16B, and Qwen3-30B-A3B, CAMEO-3b achieves up to **1.83× end-to-end speedup** over MoE-APEX in trace-driven, kernel-measured GPU-offloading evaluation and **3.64× expert-layer speedup** with up to **5.88× EDP reduction** over MoNDE in GPU–NDP simulation, with **0.14–0.63 percentage-point** average accuracy loss versus FP16.

Research Interests

My research focuses on memory-centric efficient LLM inference, with an emphasis on KV-cache management, data-movement reduction, and hardware–software co-design for memory-centric architectures. I am particularly interested in long-context and agentic LLM workloads, as well as emerging platforms such as PIM/NDP, 3D-stacked DRAM near-memory computing, and multi-chiplet systems.

Publications and Professional Service

ASPLOS'26	**Zehao Fan**, Yunzhen Liu, Garrett Gagnon, Zhenyu Liu, Yayue Hou, Hadjer Benmeziane, Kaoutar El Maghraoui, Liu Liu. "*STARC: Selective Token Access with Remapping and Clustering for Efficient LLM Decoding on PIM Systems*."
Under Review	**Zehao Fan**, Zhenyu Liu, Yunzhen Liu, Yayue Hou, Hadjer Benmeziane, Kaoutar El Maghraoui, Liu Liu. "*CAMEO: Context-Aware Mixture-of-Experts Orchestration for Inference*."
Under Review	Zhenyu Liu, Yunzhen Liu, **Zehao Fan**, Garrett Gagnon, Yayue Hou, Nan Wu, Yangwook Kang, Liu Liu. "*BEAR: Bandwidth-Efficient Adaptive Recovery of Low-Bit Quantization Errors in Mixture-of-Experts*."
Under Review	Junfeng Wu, **Zehao Fan**, Hadjer Benmeziane, Kaoutar El Maghraoui, Liu Liu, Yinan Wang. "*MaskCoFT: Masked Co-Adaptive Fine-Tuning for Memory-Efficient MoE Inference*."
Professional Service	Reviewer, *IEEE Transactions on Computers*; External Reviewer (Sub-reviewer), DAC'26; Artifact Evaluation Committee (AEC) Reviewer, ASPLOS'26.